ModelIt!

Student Activity Pack

# Can We Engineer the Climate?

Modeling Geoengineering Technologies, Carbon Capture, and Ethical Dimensions

Grade 12L2 | NGSS HS-ESS2-4 | Cause and Effect

STUDENT NAME _____ DATE _____

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# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you heard of geoengineering? What comes to mind when you think about humans deliberately modifying the global climate?**

2. **If you could deploy a giant sunshade that cooled the planet by reflecting sunlight, but it didn\'t remove any CO2 from the atmosphere, would you do it? Why or why not?**

3. **Who should get to decide whether to deploy a technology that affects the climate for all 8 billion people on Earth? One nation? A vote? Scientists?**

4. **Is it ethical to modify the climate to fix a problem caused by some nations while the side effects may harm other nations that did not cause the problem?**

# 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

- Atmospheric CO2 Concentration
- Solar Radiation Reaching Surface
- Global Mean Temperature
- Ocean Acidification Level
- Regional Climate Disruption
- Carbon Removal Rate
- Termination Risk

| EXTERNAL | INTERNAL |
|---|---|
| | |

**Explain your reasoning: Why did you sort each component the way you did?**

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Solar Shield

SRM: Active (1.5C offset) | Emissions: Continuing | CDR: None

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Carbon Removal

SRM: None | Emissions: Current | CDR: 10 Gt/year

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Integrated Solution

SRM: None | Emissions: -50% | CDR: 5 Gt/year

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Research Governance Gap** — The absence of international regulations or agreements governing geoengineering research and potential deployment — currently, any nation with sufficient resources could unilaterally deploy SRM, as no treaty prohibits it, and the lack of governance creates risks of rogue deployment or conflict

**Ozone Layer Interaction** — The potential effect of stratospheric aerosol injection on the ozone layer — sulfate aerosols that reflect sunlight also catalyze ozone-destroying reactions, potentially worsening the ozone hole and increasing UV radiation exposure, adding a health risk to the climate intervention

**Intergenerational Equity** — The ethical dimension of decisions made today that bind future generations — deploying SRM creates a commitment that must be maintained for centuries by future generations who had no voice in the decision, raising fundamental questions about whether one generation has the right to impose such obligations on its descendants

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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**The Big Question**

**If we cannot reduce emissions fast enough to prevent catastrophic warming, should humanity deliberately intervene in Earth\'s climate system through technologies like solar radiation management or atmospheric carbon removal — and who gets to decide for the entire planet?**
Use evidence from your simulation to support your answers.

1. **Why does your model show that solar radiation management is not a solution to climate change but rather a new problem layered on top of an existing one?**

2. **What is the termination shock problem and why does it make SRM deployment a multigenerational trap? How does your model demonstrate this risk?**

3. **Why is the current scale of carbon dioxide removal so inadequate compared to annual emissions? What would it take to close this gap, and is it realistic?**

4. **How does the moral hazard concept apply to geoengineering — why might having a technological fix available actually make the climate problem worse?**

5. **Who should have the authority to make decisions about deploying technologies that alter the global climate? What principles should guide this governance?**

# 8 STEM Challenge: Design a Climate Intervention Decision Framework

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## YOUR ENGINEERING MISSION

Design a governance and evaluation framework for deciding whether, when, and how to deploy climate intervention technologies, incorporating scientific risk assessment, ethical principles, and international justice considerations.

## THE SCENARIO

Global temperatures have reached 1.8 degrees Celsius above pre-industrial levels and emission reduction efforts are failing to meet targets. A coalition of wealthy nations is proposing to deploy stratospheric aerosol injection to prevent crossing the 2-degree threshold, arguing that the risk of inaction exceeds the risk of intervention. Developing nations in the tropics, whose monsoon rainfall could be disrupted by SRM, oppose deployment without their consent. Meanwhile, CDR technology has scaled to 2 gigatons per year — significant but still far below the level needed. The United Nations has commissioned your team to design a decision framework that evaluates whether geoengineering should be deployed and, if so, under what conditions, with what safeguards, and with whose consent.

### Engineering Design Thinking

1. WHAT SCIENTIFIC CRITERIA DOES YOUR FRAMEWORK USE TO EVALUATE WHETHER GEOENGINEERING DEPLOYMENT IS JUSTIFIED?

2. HOW DOES YOUR FRAMEWORK ENSURE THAT NATIONS MOST AFFECTED BY SIDE EFFECTS HAVE GENUINE DECISION-MAKING POWER, NOT JUST A VOICE?

3. WHAT SAFEGUARDS DOES YOUR FRAMEWORK INCLUDE TO PREVENT MORAL HAZARD — THE RISK THAT GEOENGINEERING AVAILABILITY REDUCES EMISSION REDUCTION URGENCY?

4. WHAT TERMINATION PROTOCOL DOES YOUR FRAMEWORK REQUIRE — HOW IS THE LONG-TERM COMMITMENT MANAGED ACROSS GENERATIONS AND CHANGING GOVERNMENTS?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Climate scientists who model geoengineering effects on Earth systems work at NASA, NOAA, universities, and national laboratories, earning $75,000-$150,000/year. Carbon capture engineers who design and operate direct air capture and geological storage systems earn $70,000-$140,000/year. Climate policy ethicists who analyze the governance, justice, and moral dimensions of climate intervention earn $60,000-$130,000/year at universities, think tanks, and international organizations.